

Seasonal Cooperative Crop Portfolios: Risk Sharing and Spatial Diversification in UK Agriculture

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Abstract

UK arable farmers face correlated yield, price, and cost shocks across counties and crops. We study whether seasonal cooperative crop allocation can reduce income volatility and member welfare dispersion relative to decentralized benchmarks. Using a county–crop–season profit panel and a reproducible CoPPer pipeline, We extend agricultural portfolio analysis to cooperative seasonal allocation, spatial diversification, and member welfare under uncertainty. We compare observed allocations, independent profit-maximizing and diversified farmers, and local, regional, and national cooperative benchmarks under historical backtests, bootstrap scenarios, and price-scenario robustness checks. Results are reported with paired hypothesis tests and bootstrap uncertainty where available. This draft skeleton uses code-generated tables and figures only; numerical claims in the text will be populated once all required outputs are synced.

1 Introduction

Seasonal crop choice exposes UK farmers to correlated agronomic and market risk. Individual diversification within a county may be limited by land,

rotation, and local agronomic constraints, while spatial diversification across counties is difficult to coordinate without institutional arrangements.

This paper asks whether *cooperative seasonal crop allocation*—with explicit risk-sharing rules and spatial diversification—can improve member welfare relative to decentralized benchmarks. Our unit of analysis is county $\times$ crop $\times$ season, and the empirical setting is UK arable agriculture for a core crop universe selected by a documented Pareto procedure.

We extend agricultural portfolio analysis to cooperative seasonal allocation, spatial diversification, and member welfare under uncertainty. The contribution is cooperative seasonal allocation, spatial diversification, and member-level welfare evaluation under uncertainty—not price forecasting, machine learning, structural supply–demand estimation, or national central planning.

We organize the paper as follows. Section 2 reviews related work. Section 3 presents the conceptual framework. Sections 4–5 describe data and methods. Sections 6–7 report main and robustness results. Section 8 discusses implications. Section 9 summarizes limitations. Section 10 concludes.

2 Related literature and research gap

Agricultural portfolio and risk-programming models study how crop mix affects expected returns and downside risk [Hardaker et al., 2015, Bradford and Kelejian, 1997]. Spatial diversification and cooperative risk pooling appear in regional risk-management and insurance literatures, but cooperative *seasonal* allocation with explicit sharing rules and member-level welfare metrics remains less developed for UK county-scale crop portfolios.

Our gap is institutional: we evaluate cooperative coordination across counties and crops under uncertainty, with benchmarks that separate independent behavior, local cooperation, regional coordination, and a national upper-bound benchmark that is not offered as policy advice.

Price forecasting and machine learning may inform scenario design but are auxiliary in this paper; main results rely on observed and resampled historical profit panels unless naive forecasts demonstrably improve scenario generation (see appendix forecast metrics).

3 Conceptual framework

Each asset is a county–crop–season profit observation. Farmers or cooperatives choose crop allocation weights subject to land availability, crop share

bounds, and feasible crop–region combinations.

3.1 Benchmark strategies

We compare six benchmarks aligned with the CoPPer charter:

- **B1** observed historical allocation (national area-share proxy where needed);
- **B2** profit-maximizing independent farmer;
- **B3** diversified independent farmer without cross-farmer coordination;
- **B4** local cooperative (primary policy-relevant model);
- **B5** regional cooperative;
- **B6** national cooperative benchmark (upper bound on spatial diversification).

3.2 Objectives and welfare

Portfolio objectives include expected profit, mean–variance trade-offs, and CVaR-based downside-risk measures. Cooperative strategies are evaluated with member-level sharing rules—individual retention, equal sharing, land-weighted sharing, risk-adjusted sharing, and compensation schemes—and inequality metrics including Gini coefficients and income dispersion.

Aggregate cooperative gains that harm identifiable members must be flagged; cooperative performance is not judged on mean profit alone.

4 Data

4.1 Crop universe

The core crop universe is selected empirically from UK arable area shares over 20-, 10-, and 5-year windows using a Pareto cumulative-share procedure documented in the pipeline report. Included crops are wheat, barley, oats, oilseed rape, potatoes, and peas/field beans. Livestock, sugar beet, and maize are excluded with documented rationales in `configs/assumptions.yaml` and the crop-universe report.

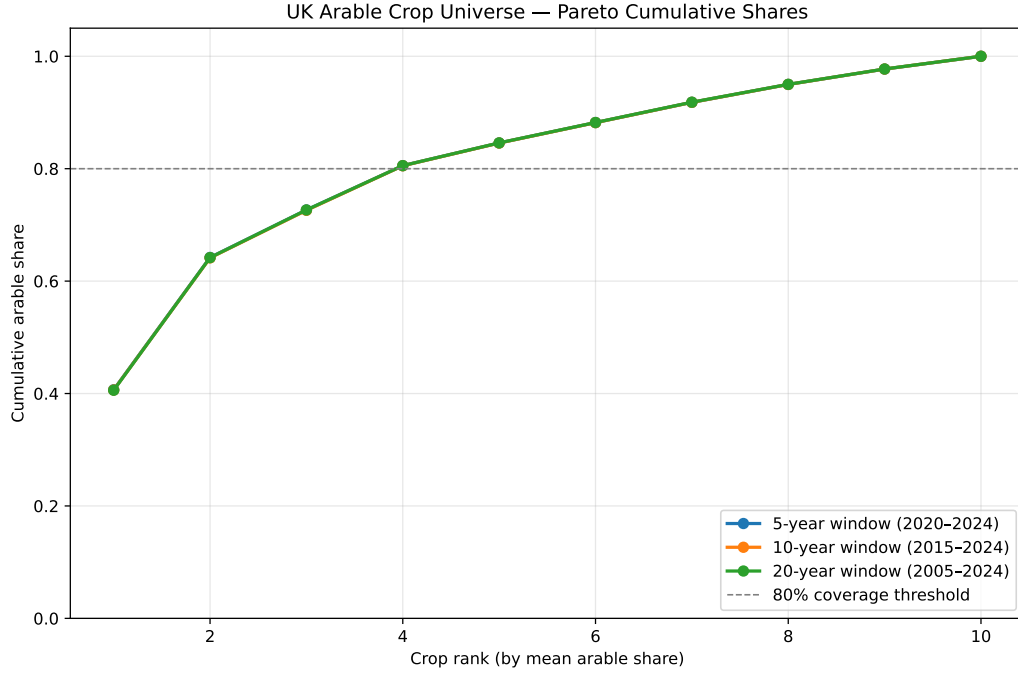


Figure 1: Pareto cumulative-share curves for UK arable crops

crop	mean_arable_share	rank	cumulative_share	window_years	year_start	year_end	meets_threshold
wheat	0.405821	1	0.405821	20	2005	2024	True
barley	0.235924	2	0.641745	20	2005	2024	True
oilseed Rape	0.084792	3	0.726537	20	2005	2024	True
fallow land	0.078880	4	0.805417	20	2005	2024	False
sugar_beet	0.040485	5	0.845902	20	2005	2024	False
other_cereals	0.036252	6	0.882153	20	2005	2024	False
peas_field_beans	0.036137	7	0.918290	20	2005	2024	False
oats	0.031730	8	0.950020	20	2005	2024	False
potatoes	0.027342	9	0.977362	20	2005	2024	False
maize	0.022638	10	1.000000	20	2005	2024	False
wheat	0.406217	1	0.406217	10	2015	2024	True
barley	0.235098	2	0.641314	10	2015	2024	True
oilseed Rape	0.084574	3	0.725889	10	2015	2024	True
fallow land	0.079314	4	0.805203	10	2015	2024	False
sugar_beet	0.040425	5	0.845627	10	2015	2024	False
other_cereals	0.036290	6	0.881917	10	2015	2024	False
peas_field_beans	0.036087	7	0.918004	10	2015	2024	False
oats	0.031859	8	0.949864	10	2015	2024	False
potatoes	0.027408	9	0.977272	10	2015	2024	False
maize	0.022728	10	1.000000	10	2015	2024	False
wheat	0.406748	1	0.406748	5	2020	2024	True
barley	0.235408	2	0.642157	5	2020	2024	True
oilseed Rape	0.084511	3	0.726667	5	2020	2024	True
fallow land	0.078787	4	0.805454	5	2020	2024	False
sugar_beet	0.040353	5	0.845807	5	2020	2024	False
other_cereals	0.036291	6	0.882098	5	2020	2024	False
peas_field_beans	0.035943	7	0.918041	5	2020	2024	False
oats	0.031900	8	0.949941	5	2020	2024	False
potatoes	0.027280	9	0.977221	5	2020	2024	False
maize	0.022779	10	1.000000	5	2020	2024	False

Table 1: Pareto crop-universe cumulative shares by window

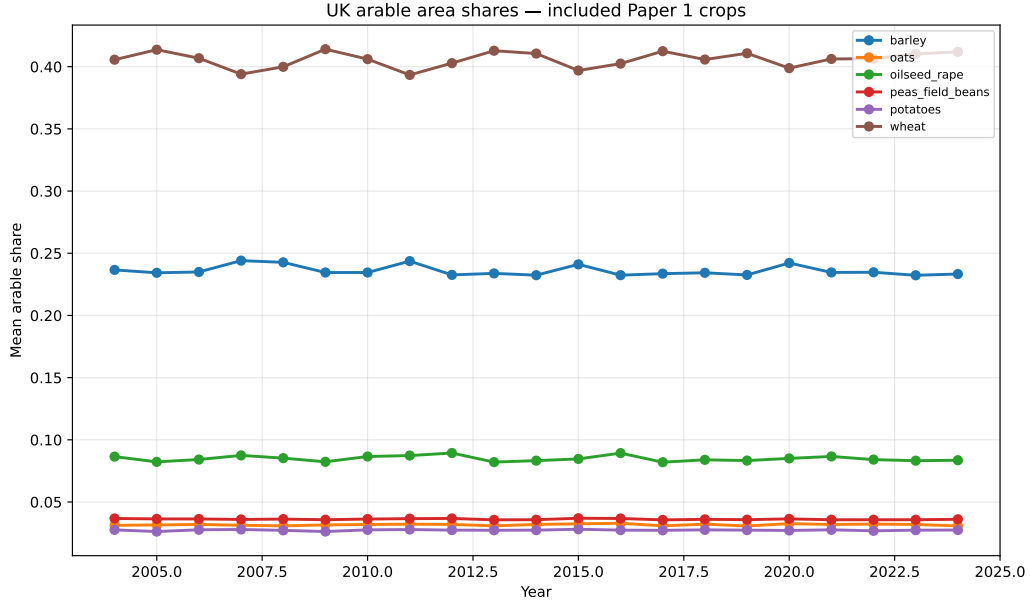


Figure 2: Crop area trends for included crops

4.2 Profit panel

County-crop-season profits follow $\text{profit} = \text{yield} \times \text{price} - \text{cost}$ with harmonized units and inflation adjustment where configured. Imputations and fallbacks are logged; no silent imputation is permitted.

crop	mean_yield	mean_price	mean_cost	mean_profit	profit_std	n_observations	share_imputed_yield
potatoes	46.518302	180.220291	2556.208266	5818.225508	455.280211	252	0.750000
oilseed rape	3.416841	368.454950	429.442989	827.559877	83.838376	252	0.750000
wheat	8.270067	169.444288	664.614149	734.197487	86.838420	252	0.750000
barley	6.710940	149.003165	593.040318	405.477284	53.547648	252	0.750000
peas_field_beans	3.614937	245.835605	490.791987	396.764584	43.089068	252	0.750000
oats	5.676456	138.680268	531.691319	254.287217	46.365761	252	0.750000

Table 2: Profit panel summary statistics by crop

Descriptive summaries by county are exported as CSV for supplementary review.

4.3 Provenance

Data sources, access dates, hashes, and processing scripts are recorded in the provenance metadata and summarized in the supplementary data-provenance report.

crop	mean_price_gbp_per_tonne	price_std	min_price	max_price	n_observations	share_imputed
barley	149.003165	12.477734	128.737289	167.432096	252	0.000000
oats	138.680268	13.052092	120.736940	159.008004	252	0.000000
oilseed Rape	368.454950	32.831469	313.155577	424.012597	252	0.000000
peas_field_beans	245.835605	18.079271	206.621827	277.128986	252	0.000000
potatoes	180.220291	12.394132	160.076868	204.093951	252	0.000000
wheat	169.444288	15.751475	142.053007	194.331453	252	0.000000

Table 3: Crop-season price summary by crop

crop	mean_cost_gbp_per_ha	cost_std	min_cost	max_cost	n_observations	share_imputed
barley	593.040318	34.498903	534.069982	646.896733	252	0.190476
oats	531.691319	30.930051	478.821363	579.976381	252	0.190476
oilseed Rape	429.442989	24.981965	386.740331	468.442462	252	0.190476
peas_field_beans	490.791987	28.550817	441.988950	535.362813	252	0.190476
potatoes	2556.208266	148.702170	2302.025783	2788.347986	252	0.190476
wheat	664.614149	38.662564	598.526703	724.970476	252	0.190476

Table 4: Crop-season cost summary by crop

5 Methods

5.1 Scenario generation

We evaluate strategies under:

1. historical backtests on observed seasons;
2. dependence-preserving Monte Carlo bootstrap resampling of the profit panel; and
3. price-scenario robustness (observed, rolling average, adverse shock, oversupply penalties).

Oversupply penalties are robustness scenarios, not identified structural price effects.

5.2 Portfolio optimization

Mean–variance and CVaR objectives are solved with documented constraints (long-only weights, maximum crop share caps, feasible crop–region sets). Covariance matrices are estimated with sample, shrinkage, and robust estimators; sensitivity is reported in robustness tables.

5.3 Inference

Hypothesis tests compare cooperative and independent benchmarks with paired tests, bootstrap confidence intervals, and effect sizes where configured. Random seeds and package versions are fixed in configuration files.

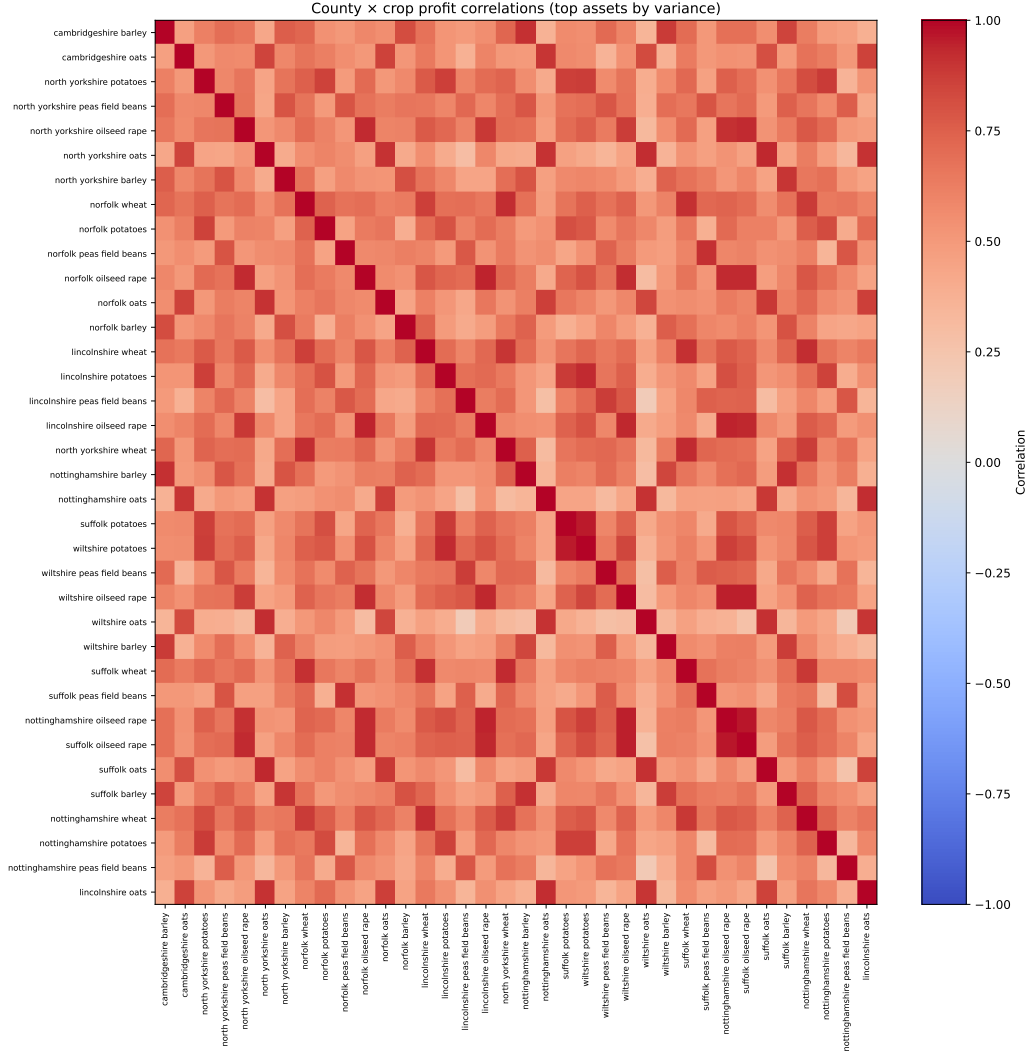


Figure 3: County $\times$ crop profit correlation heatmap (top assets by variance)

crop	barley	oats	oilseed_rape	peas_field_beans	potatoes	wheat
barley	1.000000	0.499363	0.650117	0.713168	0.598127	0.778926
oats	0.499363	1.000000	0.565235	0.479113	0.589095	0.602371
oilseed_rape	0.650117	0.565235	1.000000	0.710688	0.779720	0.772631
peas_field_beans	0.713168	0.479113	0.710688	1.000000	0.563217	0.765211
potatoes	0.598127	0.589095	0.779720	0.563217	1.000000	0.757795
wheat	0.778926	0.602371	0.772631	0.765211	0.757795	1.000000

Table 5: National crop-level profit correlation matrix

5.4 Forecasting (auxiliary)

Walk-forward price forecast evaluation (naive, rolling mean, seasonal naive, gradient boosting) supports appendix discussion only and does not enter main cooperative allocation results unless forecasts beat naive baselines.

6 Results

6.1 Benchmark comparison

Figure 4 and Table 6 summarize historical backtest performance across benchmarks B1–B6, including mean profit, volatility, CVaR, probability of loss, and worst 5% outcomes.

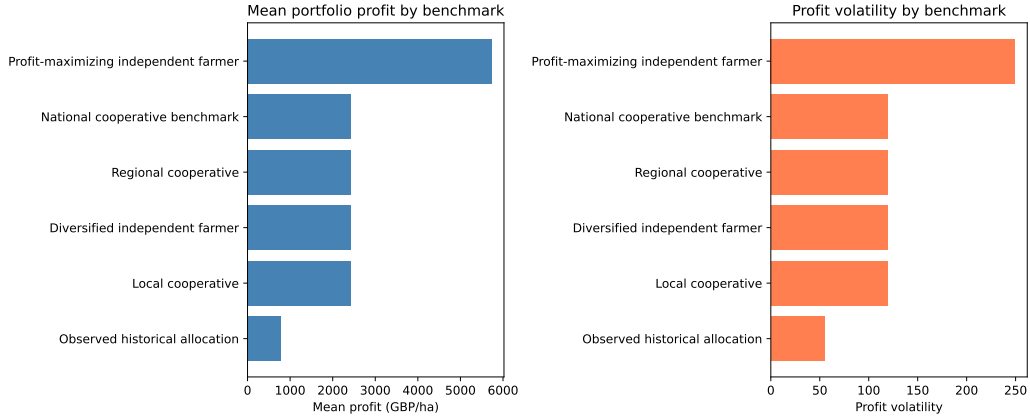


Figure 4: Historical backtest comparison across benchmarks

benchmark_id	benchmark_name	mean_profit	profit_volatility	cvar_95	probability_of_loss	worst_5pct	n_seasons
B1_observed	Observed historical allocation	774.209650	54.588309	681.893951	0.000000	699.267073	18.000000
B2_profit_maximizer	Profit-maximizing independent farmer	5335.827619	249.437795	5332.457912	0.000000	5335.702858	18.000000
B3_individual_diversified	Diversified independent farmer	2425.020647	119.092877	2228.327243	0.000000	2229.082176	18.000000
B4_local_cooperative	Local cooperative	2425.020647	119.092877	2228.327243	0.000000	2229.082176	18.000000
B5_regional_cooperative	Regional cooperative	2425.020647	119.092877	2228.327243	0.000000	2229.082176	18.000000
B6_national_cooperative	National cooperative benchmark	2425.020647	119.092877	2228.327243	0.000000	2229.082176	18.000000

Table 6: Historical backtest summary across benchmarks

6.2 Cooperative scale

Figure 5 compares local, regional, and national cooperative benchmarks.

benchmark_id	benchmark_name	mean_profit	profit_volatility	cvar_95	probability_of_loss	worst_5pct
B1_observed	Observed historical allocation	774.209650	54.588309	681.893951	0.000000	699.267073
B2_profit_maximizer	Profit-maximizing independent farmer	5735.827619	249.437795	5332.457912	0.000000	5335.702858
B3_individual_diversified	Diversified independent farmer	2425.020647	119.092877	2228.327243	0.000000	2229.082176
B4_local_cooperative	Local cooperative	2425.020647	119.092877	2228.327243	0.000000	2229.082176
B5_regional_cooperative	Regional cooperative	2425.020647	119.092877	2228.327243	0.000000	2229.082176
B6_national_cooperative	National cooperative benchmark	2425.020647	119.092877	2228.327243	0.000000	2229.082176

Table 7: CVaR and downside-risk metrics by benchmark

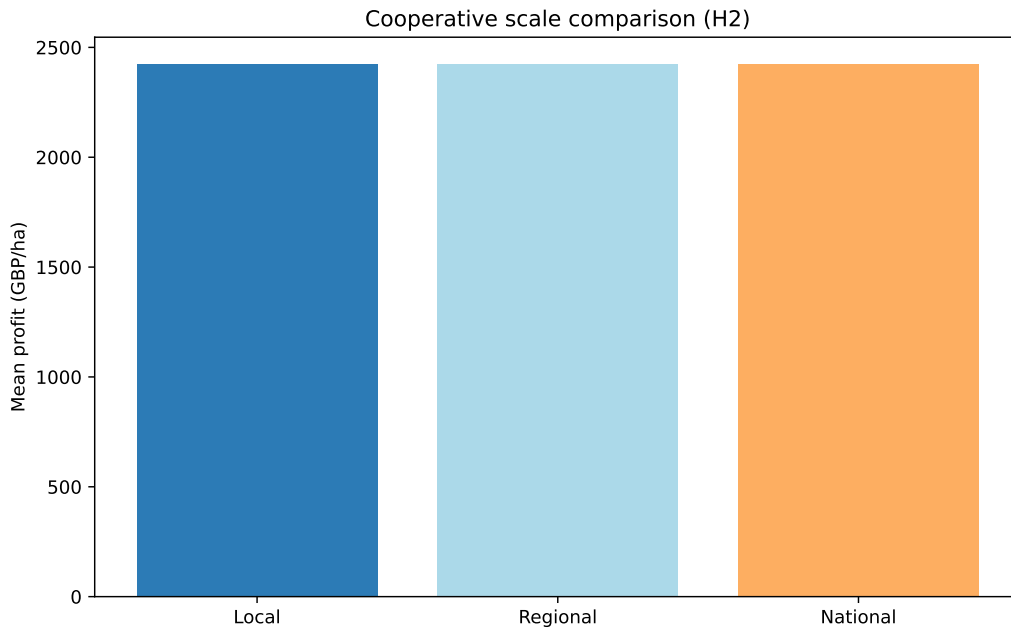


Figure 5: Cooperative scale comparison (local, regional, national)

risk_aversion	expected_return	portfolio_variance	portfolio_std
0.500000	721.571125	1825.425052	42.724993
1.000000	380.834082	803.618129	28.348159
2.000000	350.577334	746.272457	27.317988
5.000000	350.577334	746.272457	27.317988

Table 8: Mean-variance efficient frontier solutions

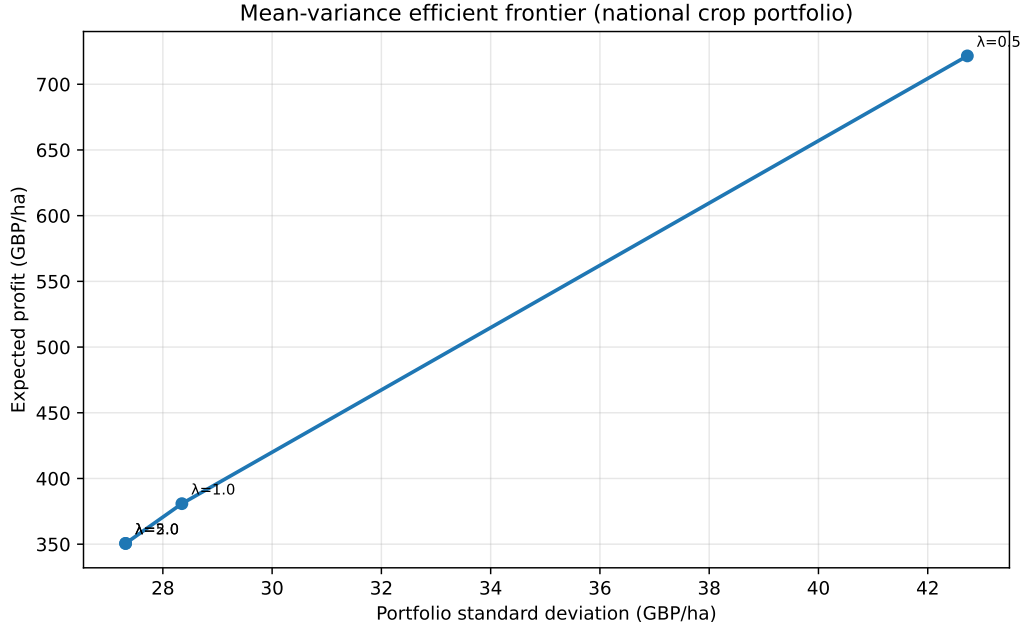


Figure 6: Efficient frontier of portfolio objectives

objective	crop	weight	expected_return	status
expected_profit	barley	0.000000	2511.768195	optimal
expected_profit	oats	0.000000	2511.768195	optimal
expected_profit	oilseed_rape	0.343333	2511.768195	optimal
expected_profit	peas_field_beans	0.000000	2511.768195	optimal
expected_profit	potatoes	0.343333	2511.768195	optimal
expected_profit	wheat	0.313333	2511.768195	optimal
mean_variance	barley	0.338154	380.834082	optimal
mean_variance	oats	0.313750	380.834082	optimal
mean_variance	oilseed_rape	0.000000	380.834082	optimal
mean_variance	peas_field_beans	0.343333	380.834082	optimal
mean_variance	potatoes	0.004764	380.834082	optimal
mean_variance	wheat	0.000000	380.834082	optimal

Table 9: Cooperative portfolio weights by objective

6.3 Efficient frontier and weights

6.4 Bootstrap simulation

Bootstrap scenario summaries are exported to supplementary CSV tables in `paper/tables/bootstrap_simulation_summary.csv`.

6.5 Welfare and sharing rules

Table 10 reports Gini coefficients, dispersion, and profit summaries by sharing rule.

sharing_rule	gini	dispersion	mean_profit	min_profit
individual	0.038008	0.070214	1268.518688	1127.710338
equal	0.000000	0.000000	1268.518688	1268.518688
land_weighted	0.000000	0.000000	1268.518688	1268.518688
risk_adjusted	0.033848	0.065983	1268.518688	1178.109018
compensation	0.038499	0.071292	5362.281223	4779.774842

Table 10: Welfare comparison across sharing rules

sharing_rule	gini_coefficient
individual	0.038008
equal	0.000000
land_weighted	0.000000
risk_adjusted	0.033848
compensation	0.038499

Table 11: Gini coefficients by sharing rule

member_id	pre	post	change	pct_change	sharing_rule
cambridgeshire	1402.636623	1402.636623	0.000000	0.000000	individual
east_riding_yorkshire	1252.577062	1252.577062	0.000000	0.000000	individual
essex	1325.834537	1325.834537	0.000000	0.000000	individual
hampshire	1133.022091	1133.022091	0.000000	0.000000	individual
herefordshire	1127.710338	1127.710338	0.000000	0.000000	individual
kent	1250.375087	1250.375087	0.000000	0.000000	individual
lincolnshire	1326.702450	1326.702450	0.000000	0.000000	individual
norfolk	1368.750815	1368.750815	0.000000	0.000000	individual
north_yorkshire	1234.566112	1234.566112	0.000000	0.000000	individual
nottinghamshire	1291.480834	1291.480834	0.000000	0.000000	individual
suffolk	1330.785741	1330.785741	0.000000	0.000000	individual
wiltshire	1177.782569	1177.782569	0.000000	0.000000	individual
cambridgeshire	1402.636623	1268.518688	-134.117935	-0.095618	equal
east_riding_yorkshire	1252.577062	1268.518688	15.941626	0.012727	equal
essex	1325.834537	1268.518688	-57.315849	-0.043230	equal
hampshire	1133.022091	1268.518688	135.496597	0.119589	equal
herefordshire	1127.710338	1268.518688	140.808350	0.124862	equal
kent	1250.375087	1268.518688	18.143601	0.014511	equal
lincolnshire	1326.702450	1268.518688	-58.183762	-0.043856	equal
norfolk	1368.750815	1268.518688	-100.232127	-0.073229	equal
north_yorkshire	1234.566112	1268.518688	33.952577	0.027502	equal
nottinghamshire	1291.480834	1268.518688	-22.962145	-0.017780	equal
suffolk	1330.785741	1268.518688	-62.267052	-0.046790	equal
wiltshire	1177.782569	1268.518688	90.736120	0.077040	equal
cambridgeshire	1402.636623	1268.518688	-134.117935	-0.095618	land_weighted
east_riding_yorkshire	1252.577062	1268.518688	15.941626	0.012727	land_weighted
essex	1325.834537	1268.518688	-57.315849	-0.043230	land_weighted
hampshire	1133.022091	1268.518688	135.496597	0.119589	land_weighted
herefordshire	1127.710338	1268.518688	140.808350	0.124862	land_weighted
kent	1250.375087	1268.518688	18.143601	0.014511	land_weighted
lincolnshire	1326.702450	1268.518688	-58.183762	-0.043856	land_weighted
norfolk	1368.750815	1268.518688	-100.232127	-0.073229	land_weighted
north_yorkshire	1234.566112	1268.518688	33.952577	0.027502	land_weighted
nottinghamshire	1291.480834	1268.518688	-22.962145	-0.017780	land_weighted
suffolk	1330.785741	1268.518688	-62.267052	-0.046790	land_weighted
wiltshire	1177.782569	1268.518688	90.736120	0.077040	land_weighted
cambridgeshire	1402.636623	1189.821781	-212.814842	-0.151725	risk_adjusted
east_riding_yorkshire	1252.577062	1228.263078	-24.313984	-0.019411	risk_adjusted
essex	1325.834537	1242.530324	-83.304213	-0.062832	risk_adjusted
hampshire	1133.022091	1379.152887	246.130796	0.217234	risk_adjusted
herefordshire	1127.710338	1449.546347	321.836008	0.285389	risk_adjusted
kent	1250.375087	1343.081836	92.706749	0.074143	risk_adjusted
lincolnshire	1326.702450	1230.181129	-96.521321	-0.072753	risk_adjusted
norfolk	1368.750815	1190.722103	-178.028713	-0.130067	risk_adjusted
north_yorkshire	1234.566112	1301.864182	67.298070	0.054512	risk_adjusted
nottinghamshire	1291.480834	1178.109018	-113.371815	-0.087784	risk_adjusted
suffolk	1330.785741	1240.044338	-90.741403	-0.068186	risk_adjusted
wiltshire	1177.782569	1248.907238	71.124669	0.060389	risk_adjusted
cambridgeshire	1402.636623	5979.390888	4576.754265	3.262965	compensation
east_riding_yorkshire	1252.577062	5229.110609	3976.533547	3.174682	compensation
essex	1325.834537	5613.135015	4287.300478	3.233662	compensation
hampshire	1133.022091	4794.661991	3661.639899	3.231746	compensation
herefordshire	1127.710338	4779.774842	3652.064503	3.238477	compensation
kent	1250.375087	5371.579022	4121.203934	3.295974	compensation
lincolnshire	1326.702450	5562.710102	4236.008351	3.192885	compensation
norfolk	1368.750815	5775.132805	4406.381989	3.219273	compensation
north_yorkshire	1234.566112	5205.419233	3970.853121	3.216396	compensation
nottinghamshire	1291.480834	5523.491969	4232.011135	3.276867	compensation
suffolk	1330.785741	5573.435952	4242.650211	3.188079	compensation
wiltshire	1177.782569	4939.531556	3761.748987	3.193925	compensation

Table 12: Member-level welfare changes by sharing rule

6.6 Hypothesis tests

benchmark_a	benchmark_b	mean_diff	ci_lower	ci_upper	p_value	cohens_d	n_seasons	hypothesis
B4_local_cooperative	B2_profit_maximizer	-3310.806971	-3374.817242	-3251.669667	0.000000	-16.939304	18	H1_local_vs_independent
B6_national_cooperative	B4_local_cooperative	0.000000	0.000000	0.000000	0.022926	0.000000	18	H2_national_vs_local
B5_regional_cooperative	B4_local_cooperative	0.000000	-0.000000	0.000000	0.298021	0.000000	18	H2_regional_vs_local
B3_individual_diversified	B2_profit_maximizer	-3310.806971	-3374.817242	-3251.669667	0.000000	-16.939304	18	H1_diversified_vs_independent

Table 13: Hypothesis test summary for cooperative benchmarks

7 Robustness checks

Robustness dimensions follow charter §16: historical windows, crop-universe variants, cooperative scales, price assumptions, risk-aversion parameters, scenario-generation methods, covariance estimators, and cost/yield fallback assumptions where implemented.

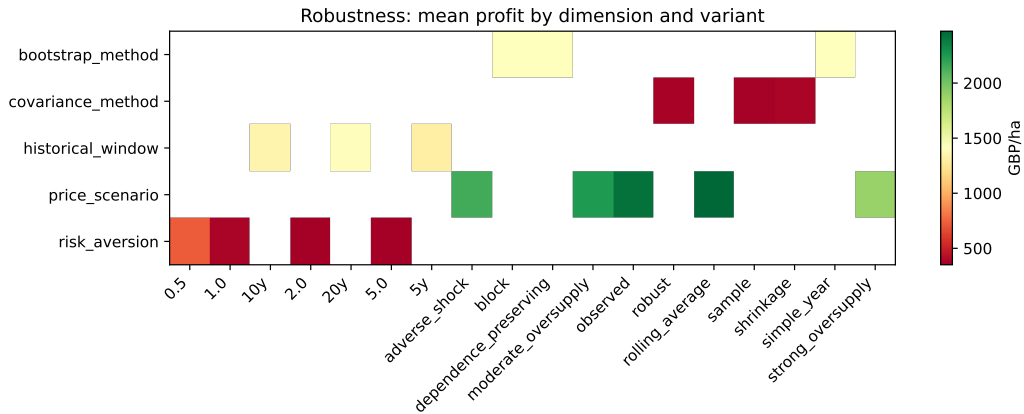


Figure 7: Robustness heatmap across configured dimensions

Price-scenario and covariance-estimator sensitivity appear in the robustness summary table. Crop-universe robustness variants and dedicated historical-window tables are reported where generated by `scripts/run_robustness.py`; see supplementary markdown reports for full audit trails.

8 Discussion

Cooperative seasonal allocation may reduce volatility relative to independent profit-maximizing behavior when spatial diversification and sharing rules

dimension	variant	mean_profit	profit_volatility	cvar_95	opt_expected_return	expected_return	portfolio_std	portfolio_variance
price_scenario	observed	2425.020647	119.092877	2228.327243	380.834082	NaN	NaN	NaN
price_scenario	rolling_average	2470.886194	109.181197	2287.216628	402.468349	NaN	NaN	NaN
price_scenario	adverse_shock	2144.675636	122.964828	1941.063193	276.791181	NaN	NaN	NaN
price_scenario	moderate_oversupply	2243.772125	110.290430	2062.122668	378.548157	NaN	NaN	NaN
price_scenario	strong_oversupply	1881.275082	92.755027	1728.551491	370.694890	NaN	NaN	NaN
covariance_method	sample	364.295962	NaN	NaN	NaN	364.295962	28.128276	NaN
covariance_method	shrinkage	380.834082	NaN	NaN	NaN	380.834082	28.348159	NaN
covariance_method	robust	369.481504	NaN	NaN	NaN	369.481504	28.591103	NaN
risk_aversion	0.500000	721.571125	NaN	NaN	NaN	721.571125	NaN	1825.425052
risk_aversion	1.000000	380.834082	NaN	NaN	NaN	380.834082	NaN	803.618129
risk_aversion	2.000000	350.577334	NaN	NaN	NaN	350.577334	NaN	746.272457
risk_aversion	5.000000	350.577334	NaN	NaN	NaN	350.577334	NaN	746.272457
bootstrap_method	dependence_preserving	1406.694103	16.789540	NaN	NaN	NaN	NaN	NaN
bootstrap_method	simple_year	1406.694103	16.789540	NaN	NaN	NaN	NaN	NaN
bootstrap_method	block	1404.926290	25.533764	NaN	NaN	NaN	NaN	NaN
historical_window	20y	1397.507114	75.494781	NaN	NaN	NaN	NaN	NaN
historical_window	10y	1342.619288	55.094193	NaN	NaN	NaN	NaN	NaN
historical_window	5y	1305.341211	40.994183	NaN	NaN	NaN	NaN	NaN

Table 14: Robustness summary across dimensions and variants

align member incentives. Regional and national benchmarks primarily illustrate upper bounds on diversification rather than practical policy prescriptions.

Welfare comparisons show that sharing rules matter for inequality metrics: equal and land-weighted rules collapse dispersion in the reference run, while individual retention and compensation schemes preserve or amplify dispersion depending on deviation from individually optimal crops.

External validity is limited to the UK arable case study, reference cost and price series, and cooperative adoption constraints not modeled here. Forecasting results remain appendix material unless they improve scenario generation.

9 Limitations

This section summarizes scientifically relevant limitations tracked in `paper/MANUSCRIPT_LIMITATIONS`; the full integration log is maintained separately.

- **Yield coverage.** County-level yield series rely on reference tables and documented national fallbacks where county data are unavailable.
- **Cost aggregation.** Enterprise gross margins and crop-level average variable costs may not capture farm-specific heterogeneity.
- **Price proxies.** Crop-season prices use published reference series; potato and minor-crop proxies may misstate local market conditions.
- **Covariance instability.** Sample covariance estimates for many county–crop assets are noisy; shrinkage and robust estimators are reported but uncertainty remains.

- **Cooperative adoption.** The model evaluates counterfactual allocations and sharing rules, not transaction costs, governance, or compliance with cooperative law.
- **UK external validity.** Institutional and agronomic conclusions may not transfer to other countries or to livestock and contract crops excluded from Paper 1.
- **Journal template.** Draft builds use an `article.cls` fallback until a frozen target journal and official publisher template are confirmed.

10 Conclusion

We present a reproducible framework for evaluating cooperative seasonal crop allocation under uncertainty in UK agriculture. We extend agricultural portfolio analysis to cooperative seasonal allocation, spatial diversification, and member welfare under uncertainty. Relative to decentralized benchmarks, cooperative strategies can alter volatility, downside risk, and member welfare depending on organizational scale and sharing rules.

Future work may tighten data linkages for county yields and costs, expand agronomic feasibility constraints, and assess adoption frictions. The manuscript layer remains a draft automation scaffold until all required outputs are synced, LaTeX compiles locally, and a target journal is frozen.

11 Reproducibility and data availability statement

The analysis is designed to be reproducible from repository scripts under the MIT license. All figures and tables intended for the paper are generated by the CoPPer pipeline and synced into `paper/figures` and `paper/tables` via `make paper-assets`. Data provenance is recorded in `data/metadata/` and summarized in supplementary reports. Random seeds are configured in `configs/scenarios.yaml` and related configuration files. Known limitations and missing datasets are logged in `paper/MANUSCRIPT_LIMITATIONS.md`.

Public release of underlying third-party datasets depends on source licenses (DEFRA, AHDB, and reference tables documented in `data/metadata/sources.csv`). We do **not** claim full reproducibility until `make reproduce` and `make paper-pdf` succeed on a clean environment with the recorded package versions.

Code, configuration, experiment registry entries, and run metadata are available in the main CoPPer repository; the Overleaf-oriented submission package excludes raw and processed data by design.

References

- Gerald Bradford and Harry H. Kelejian. Estimation of dynamic rural sector income distributions with endogenous farm numbers: An application to the north central region. *American Journal of Agricultural Economics*, 79 (3):908–919, 1997. TODO: verify DOI and page range.
- J. Brian Hardaker, Gudbrand Lien, Jock R. Anderson, and Ruud B. M. Huirne. *Coping with Risk in Agriculture: Applied Decision Analysis*. CABI, 3 edition, 2015. TODO: verify edition and publisher metadata before submission.

A Data sources and provenance

Primary data categories include UK crop area and yield statistics (DEFRA / GOV.UK), crop prices (AHDB and documented proxies), enterprise costs (Farm Business Survey and reference margin tables), and spatial county definitions.

Full source URLs, access dates, file hashes, and processing scripts are listed in `data/metadata/sources.csv` and `outputs/reports/data_provenance.md`. The synced supplementary copy is `supplementary/data_provenance.md` in the submission package.

Crop inclusion and exclusion rationales appear in `configs/assumptions.yaml` and `supplementary/crop_universe_selection.md`.

B Additional robustness and forecasting

Price forecast walk-forward metrics and the best-model summary are exported for appendix discussion only.

See `outputs/reports/robustness_forecasting.md` for the full robustness audit trail.

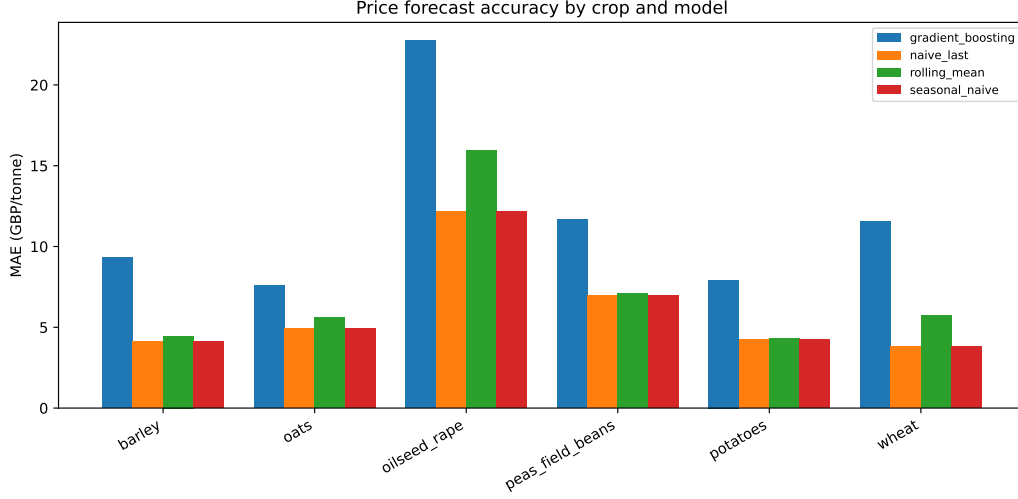


Figure 8: Price forecast accuracy (auxiliary analysis)

crop	model	mae	rmse	mape	directional_accuracy	n_forecasts
barley	naive_last	4.143357	5.229841	0.029388	0.466667	16
barley	rolling_mean	4.463446	5.723696	0.031960	0.666667	16
barley	seasonal_naive	4.143357	5.229841	0.029388	0.466667	16
barley	gradient_boosting	9.303224	10.582915	0.065167	0.733333	16
oats	naive_last	4.901768	6.285297	0.036679	0.466667	16
oats	rolling_mean	5.588374	6.642214	0.042159	0.533333	16
oats	seasonal_naive	4.901768	6.285297	0.036679	0.466667	16
oats	gradient_boosting	7.607931	9.177242	0.056670	0.666667	16
oilseed rape	naive_last	12.183862	16.297776	0.035174	0.600000	16
oilseed rape	rolling_mean	15.919659	18.520066	0.045926	0.466667	16
oilseed rape	seasonal_naive	12.183862	16.297776	0.035174	0.600000	16
oilseed rape	gradient_boosting	22.733987	26.319145	0.064433	0.733333	16
peas field beans	naive_last	6.952517	8.652075	0.029822	0.466667	16
peas field beans	rolling_mean	7.116834	9.190624	0.030600	0.600000	16
peas field beans	seasonal_naive	6.952517	8.652075	0.029822	0.466667	16
peas field beans	gradient_boosting	11.652672	14.191403	0.049493	0.666667	16
potatoes	naive_last	4.243881	5.532576	0.024464	0.400000	16
potatoes	rolling_mean	4.307566	5.256303	0.024921	0.533333	16
potatoes	seasonal_naive	4.243881	5.532576	0.024464	0.400000	16
potatoes	gradient_boosting	7.930873	9.158579	0.045654	0.666667	16
wheat	naive_last	3.786490	4.933593	0.023584	0.866667	16
wheat	rolling_mean	5.728416	6.751728	0.035763	0.866667	16
wheat	seasonal_naive	3.786490	4.933593	0.023584	0.866667	16
wheat	gradient_boosting	11.550070	13.134375	0.071120	0.933333	16

Table 15: Price forecast accuracy metrics by crop and model

crop	best_model	best_mae	naive_mae	beats_naive
barley	naive_last	4.143357	4.143357	False
oats	naive_last	4.901768	4.901768	False
oilseed_rape	naive_last	12.183862	12.183862	False
peas_field_beans	naive_last	6.952517	6.952517	False
potatoes	naive_last	4.243881	4.243881	False
wheat	naive_last	3.786490	3.786490	False

Table 16: Best price forecast model by crop

C Algorithmic details

Portfolio problems use `scipy` and `cvxpy` solvers with logged failures. CVaR optimization follows a Rockafellar–Uryasev formulation; failures on small samples are logged and non-fatal.

Bootstrap resampling defaults to dependence-preserving draws that retain cross-crop and cross-county correlation within sampled years.

Experiment IDs, configuration paths, and output locations are listed in `experiments/experiment_registry.yaml` (synced to supplementary materials).

D Integration limitations log

The living markdown file `MANUSCRIPT_LIMITATIONS.md` records missing outputs, template gaps, Overleaf workflow constraints, and blocking issues for the manuscript layer. Scientific limitations are summarized in Section 9; this appendix points reviewers and collaborators to the full integration log for build and submission readiness.

The manuscript is labeled a **draft automation scaffold** until:

- all required results exist and are synced;
- `make paper-pdf` compiles locally;
- the target journal is frozen; and
- every figure and table is traceable to code-generated outputs.